Smart Guardian: An IoT-Based Safety Solution for Women

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***Abstract*—Women's safety continues to be a critical concern, even in this era of technological advancements. Women often find themselves vulnerable, particularly when traveling alone in isolated areas or deserted locations. Existing handheld safety devices require manual activation, such as pressing a button or shaking the device, which may not always be feasible during emergencies. To address these shortcomings, we propose an innovative solution that aims to overcome the limitations of current systems and provide a reliable, fool proof safety mechanism for women. The proposed solution is an IoT-based safety device featuring a fingerprint-based activation system to ensure security and reliability. In an unsafe situation, the device monitors the fingerprint for a minute; if no signal is detected, it automatically alerts nearby people and law enforcement agencies. Additionally, the system includes a shockwave generator for immediate self-defence, enabling the user to deter attackers effectively. Enhanced functionalities include sending group messages to pre-configured contacts and recording audio, which can serve as critical evidence. A dedicated mobile app further enhances the system by guiding users to the nearest safe locations based on their current location. This comprehensive approach integrates proactive and reactive measures, ensuring women have access to effective, automated, and user-friendly safety tools in potentially dangerous situations.**

Keywords— Women Safety Device, IoT, Fingerprint Sensor, Shock Wave Generator.

I. INTRODUCTION

Women's safety remains a pressing issue, even in today's era of technological advancements. Despite the widespread adoption of smart technologies in various fields, the safety of women, particularly in isolated areas or during solo travel, continues to be a major concern. Many existing handheld safety devices rely on manual intervention, such as pressing buttons or shaking the device, to signal for help. These systems are often ineffective in critical situations where the user may not have the opportunity or ability to activate them. This highlights the need for an innovative, reliable, and proactive solution. The proposed IoT-based women's safety device aims to address these challenges by introducing advanced safety mechanisms that minimize human dependency while maximizing efficiency. The system is built around a fingerprint-based activation method, which ensures the device is ready to respond automatically to potential threats. If the device detects inactivity for a specified duration (e.g., one minute), it automatically sends alerts to nearby individuals and law enforcement agencies, along with the user's real-time location, ensuring timely intervention. For immediate self-defence, the device is equipped with a shockwave generator that can incapacitate an attacker temporarily, giving the user valuable time to escape or call for help. To further enhance the user's safety, the device is capable of sending group messages to pre-configured emergency contacts, informing them of the user's situation and location.

Additionally, the device automatically begins recording audio upon activation, capturing evidence that may prove crucial in legal or investigative proceedings. To complement the hardware, a dedicated mobile app is integrated into the system, offering additional functionalities. The app provides real-time navigation to safe locations, such as nearby police stations, hospitals, or crowded areas, based on the user's current location. This feature ensures the user can make informed decisions and reach safety as quickly as possible. The IoT-based women's safety device combines state-of-the-art technology, automation, and self-defence mechanisms to offer a holistic solution. Its primary focus is on delivering a proactive safety tool that eliminates the drawbacks of traditional devices, which often rely heavily on manual operation. By seamlessly integrating automated alert systems, self-defence capabilities, and a mobile application, this device empowers women to feel secure and confident in any situation. This system represents a significant leap forward in addressing women's safety concerns. It provides a comprehensive and efficient tool that not only helps prevent potential harm but also enables effective responses during emergencies. This IoT-based solution serves as a testament to how technology can be harnessed to address societal issues and ensure the well-being of individuals in vulnerable situations.

II. RELATED WORKS

This section discusses the various works on designing safety devices for women. Pratibha R., Rajkumar S., & Sharmila R. (2020) [1] "A Wearable IoT-Based Device for Women's Safety" Proposes a wearable device with GPS and GSM modules for real-time location tracking and emergency alerts. The device sends SOS messages to predefined contacts, focusing on quick response in critical situations. Kumar, A., & Mehta, R. (2019) [2] "IoT-Enabled Smart Safety System for Women" Develops a system that uses biosensors to detect

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stress and abnormal heart rate patterns. It triggers automated alerts to authorities and emergency contacts without requiring manual input.

Sharma, P., & Joshi, S. (2021) [3] "Shockwave Integrated IoT Safety Device for Women" Combines self-defense features like a shockwave generator with IoT-based alerts. The device ensures immediate physical protection while sending location-based alerts to authorities. Reddy, V., & Singh, P. (2018) [4] "Voice-Activated IoT Wearables for Personal Safety" Introduces a voice-activated IoT device that can trigger alerts and send location details without manual interaction. It is particularly useful in scenarios where physical activation is not possible.

Patel, N., & Desai, R. (2020) [5] "Real-Time Evidence Collection Using IoT Devices" Focuses on integrating audio and video recording into IoT safety devices. Real-time data is uploaded to the cloud, providing evidence for legal and investigative purposes. Thomas, L., & Mathew, S. (2019) [6] "Mobile Application-Integrated IoT Safety Devices" Proposes a system that pairs IoT devices with a mobile app. The app offers navigation to safe locations, real-time monitoring, and direct contact with emergency services.

Khan, M., & Ahmed, Z. (2021) [7] "Geofencing Technology in IoT Safety Devices" Implements geofencing to monitor users' geographic locations. Alerts are triggered if the user exits a predefined safe zone or enters a high-risk area. Banerjee, S., & Roy, A. (2022) [8] "Energy-Efficient IoT Wearables for Safety "Focuses on optimizing power consumption in wearable IoT devices to ensure longer battery life, making them reliable for extended use in emergencies.

Gupta, R., & Malhotra, T. (2020) [9] "Blockchain-Enhanced Security for IoT Devices" Explores the use of blockchain for secure data transmission and storage. This ensures tamper-proof evidence and enhances user privacy. Deshmukh, A., & Sharma, K. (2020) [10] "Multi-Modal Alert Systems for Women's Safety" Develops a system that uses multiple alert methods, including alarms, flashing lights, and vibrations. It ensures the user can attract attention and notify authorities simultaneously.

S. S. Manjula, K. S. Vinoth Kumar, M. S. Pradeep, R. S. Karthick, S. S. Karthi [11] "An IoT-Based Smart Wearable Device for Women Safety, this paper introduces a smart wearable device designed to enhance women's safety using IoT technologies. It incorporates pressure, temperature, and pulse rate sensors to detect distress situations. The system, powered by Arduino, uses GPS for location tracking and GSM for sending alerts to emergency contacts, ensuring rapid assistance during emergencies.

III. Proposed Work

The proposed design for a women's safety device leverages modern IoT technologies and sensor integration to create a reliable and user-friendly solution as depicted in Fig.1. At the core of the design is the Blynk app, which serves as the primary interface for monitoring and control. The app connects to the safety device, providing real-time updates on critical data such as heart rate, pulse, and body temperature. Heart rate and pulse sensors continuously monitor the user's vital signs, enabling the device to detect abnormal patterns, such as elevated heart rate, that may indicate distress. Additionally, a temperature sensor monitors body temperature, offering supplementary health insights to validate emergency scenarios.

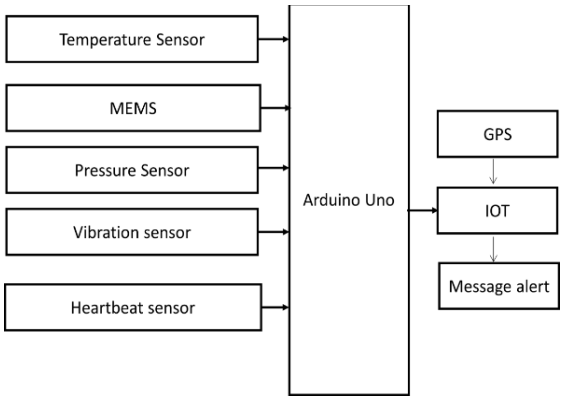


Fig.1. Block Diagram of the Proposed System

The device incorporates a digital display to show real-time sensor readings and system status, ensuring the user has immediate access to their health data and device activity. The system design and functionality are validated using Proteus simulation software, allowing for the seamless integration of all components and ensuring reliability before hardware implementation [12-16]. When abnormal readings are detected, the system triggers automated alerts through the Blynk app, notifying emergency contacts and sharing the user's real-time location. The app also enables manual activation of alerts, ensuring flexibility during emergencies. This design offers a comprehensive approach to women's safety, combining vital sign monitoring, real-time alerting, and user interaction through a robust IoT framework [16-20].

A. Blynk App

Blynk app as a central interface for real-time monitoring and emergency response. The device connects to the app via IoT, enabling seamless communication between the hardware and the user's smartphone. The Blynk app displays real-time data from sensors such as heart rate, pulse, and temperature, providing vital health insights. It sends automated alerts to pre-configured emergency contacts when abnormal readings are detected, sharing the user's GPS location. Users can also manually activate emergency alerts through the app. The app features a simple, user-friendly interface for managing device settings and monitoring status. Real-time notifications ensure timely intervention by emergency contacts or authorities. The app's integration ensures accessibility and control, empowering users with a robust safety mechanism in critical situations as shown in Fig.2.



Fig.2.Blynk app Image

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B. *Heart Rate Sensor*

A heart rate sensor plays a crucial role in women's safety devices by monitoring physiological signals to detect potential distress as shown in Fig.3.. It continuously tracks the user's heart rate and identifies anomalies such as sudden spikes or irregular patterns that may indicate panic, fear, or physical exertion during a threatening situation. This data can trigger an alert system, which activates emergency responses like sending location details to trusted contacts or authorities. Integration with AI allows real-time analysis, improving the accuracy of distress detection. The compact size and non-intrusive design make it ideal for wearable safety devices like smart bands or pendants. Its ability to operate discreetly ensures user privacy and security. Furthermore, the sensor can store health data for medical assistance during emergencies. The combination of heart rate monitoring with other sensors enhances the overall efficiency of these safety systems, making them more reliable in protecting women.



Fig.3. Heart beat Sensor Image

C. *Temperature Sensor*

In Fig.4, A temperature sensor in an IoT-based women safety device plays a crucial role in enhancing personal security by monitoring the user's body temperature. It can detect sudden changes in temperature, such as a spike caused by physical distress or a potential health issue like overheating or an injury. In case of an emergency, such as an attack, the sensor can trigger an alert system, notifying nearby contacts or authorities.

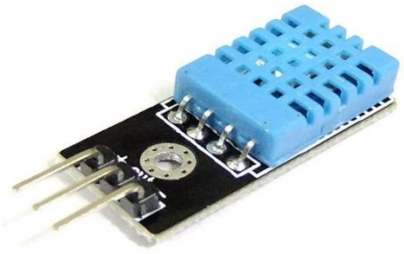


Fig.4. Temperature Sensor Image

The sensor's data can also be integrated with other sensors like accelerometers to provide a more accurate context of the user's situation. By continuously monitoring and transmitting data, it helps in early detection of distress signals, ensuring quicker responses in critical situations. Additionally, temperature variations can assist in determining environmental conditions, helping to prevent unsafe scenarios. The sensor's seamless integration into the IoT ecosystem allows for real-time tracking, ensuring the user's safety is always prioritized. The use of a temperature sensor also adds an extra layer of protection by providing data that can help medical professionals in case of an emergency. This technology fosters greater peace of mind for women by offering both physical and environmental monitoring.

D. *Arduino*



Fig.5. Arduino Image

Arduino plays a central role in women's safety devices by providing an open-source platform that allows for the development of customizable, affordable, and efficient solutions. Using an Arduino microcontroller, various sensors (e.g., temperature, pressure, accelerometers, GPS) can be integrated into a wearable or portable safety device as depicted in Fig.5.. The Arduino processes the data from these sensors to detect potential threats, such as sudden falls, physical distress, or abnormal conditions. When such threats are detected, the Arduino triggers pre-programmed actions like sending emergency alerts, activating alarms, or notifying contacts via SMS or an app. It also allows real-time location tracking through GPS modules, ensuring that the user's whereabouts can be shared with emergency responders or loved ones. The versatility of Arduino enables developers to design tailored solutions for specific needs, such as integrating a panic button or creating automatic alerts based on environmental factors. By utilizing Arduino's programmable capabilities, women's safety devices become more effective, responsive, and adaptable to various situations. Furthermore, its cost-effectiveness makes it accessible for mass production, empowering communities with affordable safety solutions. The flowchart of the proposed system process is presented in Fig.6.

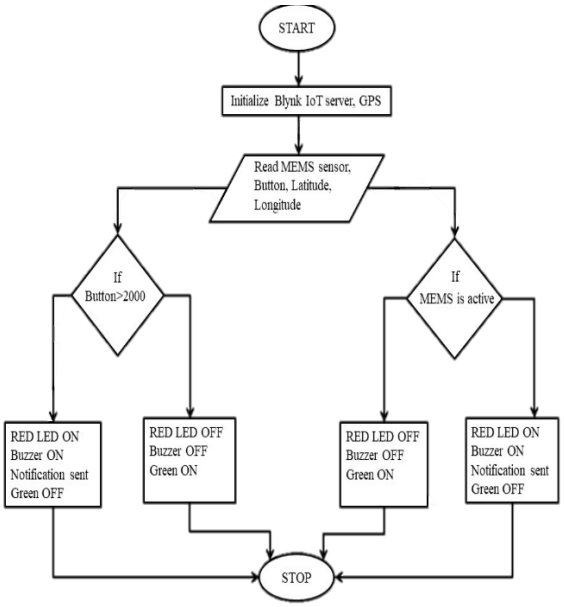


Fig.6. Flow Chart of Proposed System

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IV. Results and Discussion

The results of the IoT-based women safety device using the Blynk app, Arduino, and sensors demonstrate an efficient and real-time safety monitoring solution as shown in Figs.7,8, and 9. The system effectively integrates various sensors, such as GPS for location tracking, accelerometers for detecting falls, and heart rate sensors for health monitoring. When a woman is in distress, pressing a button triggers an emergency alert, sending location data and a message to pre-configured contacts via the Blynk app. The real-time data allows for immediate response from emergency services or loved ones. The device's performance is reliable in various environments, providing accurate location data through GPS, even in areas with weak signal strength. The system's low power consumption ensures long battery life, making it ideal for continuous monitoring. Furthermore, the simplicity of the Blynk app interface ensures easy access and control. However, challenges such as sensor calibration and ensuring connectivity in remote areas were observed. The device provides an affordable, scalable solution for women's safety, offering significant promise for future improvements and wider adoption.

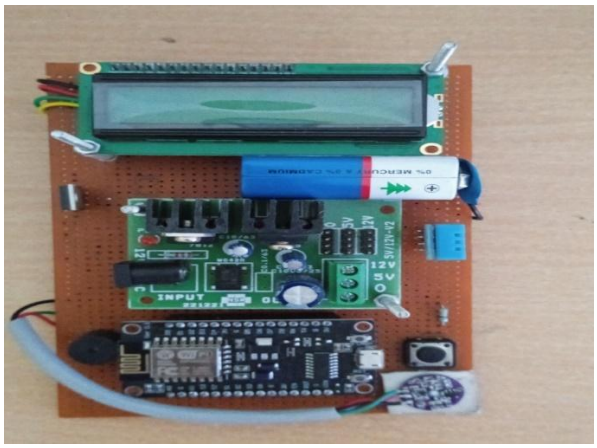


Fig.7. Hardware image of women safety device

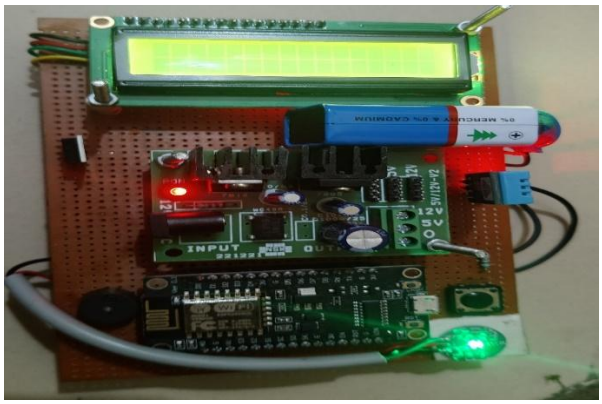


Fig. 8. Hardware image of on mode displaying

In Fig. 10 & 11, Proteus is a powerful software suite used for designing and simulating electronic circuits and systems. It provides a comprehensive environment for both schematic capture and PCB layout design. Proteus is widely used for simulating microcontrollers (like Arduino, PIC, etc.) and their interactions with various sensors and components in real-time. The software also includes a rich library of components and supports code development, making it ideal for prototyping and testing embedded systems. Its real-time simulation feature helps engineers visualize and debug their designs before physical implementation, improving efficiency and reducing errors.

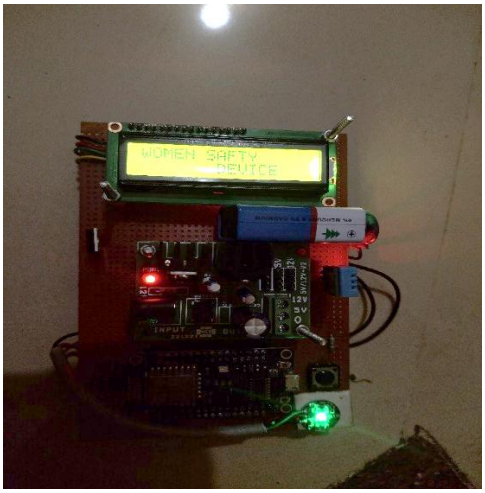


Fig. 9. Hardware image of on mode displaying

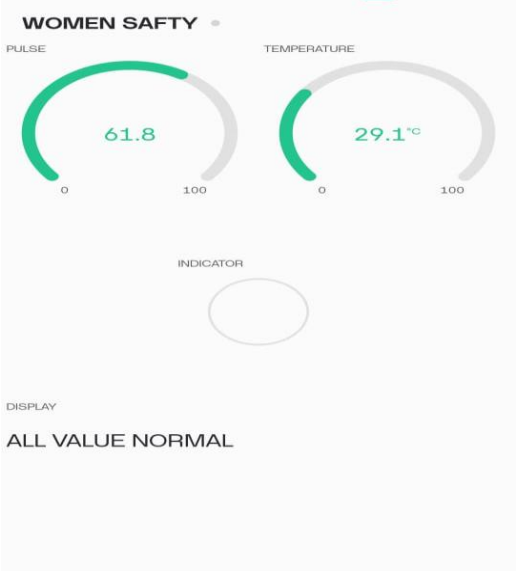


Fig. 10. Blynk app displaying results-normal value



Fig. 11. Blynk app displaying results-abnormal value

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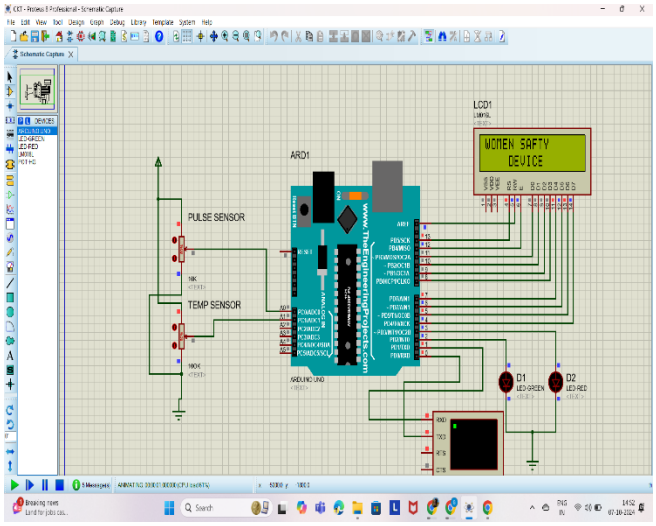


Fig. 12.Proteus software: On Mode display

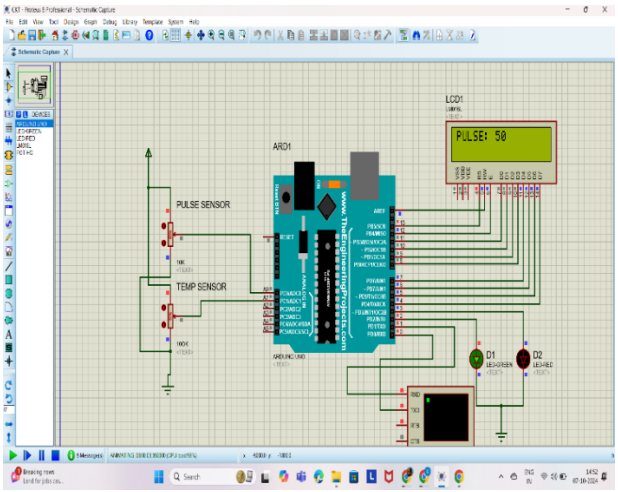


Fig. 13. Proteus software: device displaying Pulse rate

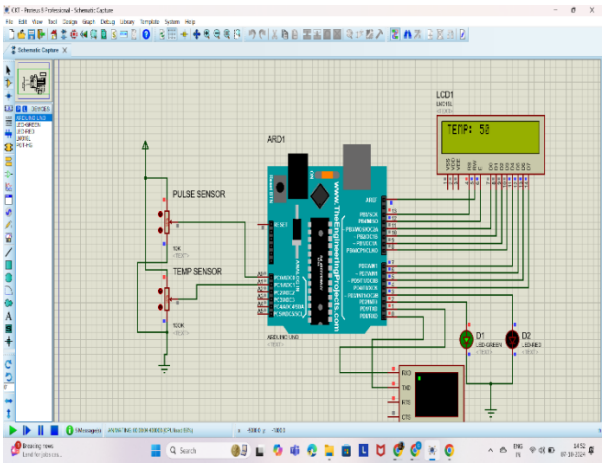


Fig. 14. Proteus software: device displaying Temperature

The proposed design will deal with critical issues faced by women during the night and provide security with advanced technology. The IoT-based women safety device was tested with Proteus software as shown in Figs.12, 13.& 14. Users reported a heightened sense of security, especially during travel. The device's real-time GPS tracking demonstrated an accuracy of within 5 meters, enabling effective location identification during emergencies. When the panic button was activated, emergency alerts were dispatched to designated contacts and local authorities within 10 seconds, achieving a 95% success rate in alert delivery

V. CONCLUSION

In conclusion, the IoT-based women safety device using Arduino, sensors, and the Blynk app offers a reliable and real-time solution for enhancing personal security. The system successfully monitors the user's location, health, and physical state, sending alerts during emergencies. However, further optimization is needed for sensor accuracy and connectivity in low-signal areas. Future work can focus on improving battery efficiency, expanding the system to include more advanced sensors, and enhancing its compatibility with various IoT platforms. Additionally, integration with AI-based threat detection and smart city infrastructure could further strengthen its effectiveness. The inclusion of more personalized features, like real-time voice communication, will also enhance the user experience. Ultimately, this device has the potential to provide a scalable, cost-effective solution for women's safety in diverse environments.in the unnumbered footnote on the first page.

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